

Stakeholder Engagement Plan

1. Project Context

Customer support systems are frequently overwhelmed by high volumes of unstructured text, making manual complaint triage slow, error-prone, and costly. This project develops an end-to-end automated complaint classification pipeline using a transformer-based architecture, fine-tuned to handle colloquial language, spelling errors, and class imbalance in real-world consumer complaints — including Nigerian English and Pidgin phrasing. Because the system directly mediates how complaints are read and routed, its design and evaluation cannot be left to the engineering team alone; it requires deliberate engagement with the people who are affected by it and the people who govern it.

2. Stakeholder Classification

Two stakeholder groups were selected as most vital to this project, representing the two poles of the system: the end user experiencing the problem, and the decision-maker funding and judging the solution.

Stakeholder Group	Category	Description
Customers / Complainants	Community members (end users)	Individuals submitting complaints that the model must read, interpret, and route correctly.
Business / Support Leadership	Institutional decision-makers / funders	Those funding, deploying, and judging the system against efficiency and reliability targets.

3. Interests and Influence

Top-ranked interest per group

Customers/Complainants — Accuracy of understanding. The most direct harm from this system is a misread complaint: if the model misinterprets colloquial or Pidgin phrasing, the customer's issue is misrouted or ignored. This is prioritized above speed or fairness because it is the failure customers would notice and be angriest about first.

Business/Support Leadership — Efficiency gains. The system is being funded specifically to reduce manual triage time and cost. Model accuracy matters to this group too, but instrumentally — as the means of trusting the system enough to deploy it, not as an end in itself.

Influence ranking

Customers/Complainants — Low influence. They cannot redesign the model, set evaluation criteria, or alter the deployment timeline. Their only leverage is indirect and slow: misrouted complaints accumulate into visible metrics or public complaints that leadership eventually notices.

Business/Support Leadership — High influence. This group controls the budget, defines the success thresholds (e.g. precision/recall/F1 targets), and holds the go/no-go decision on deployment.

4. Interest vs. Influence Matrix

Both groups carry high interest in the project's outcome; what separates them is influence. This asymmetry — the people most affected by the system having the least power over how it is built — is a recurring concern in participatory AI literature (Birhane et al., 2022; Wakunuma & Jiya, 2019) and is a deliberate consideration in this engagement plan rather than an oversight.

	Low Influence	High Influence
High Interest	Customers / Complainants Strategy: Keep Informed	Business / Support Leadership Strategy: Manage Closely

5. Engagement Strategies

Customers / Complainants — Keep Informed

Effort here is transparency-focused rather than decision-sharing, since this group cannot directly alter system design but is deeply affected by its outputs. Concrete actions:

- Provide a clear, visible acknowledgment when a complaint is received and routed, including an easy path to flag a wrong categorization.
- Publish plain-language notices (not technical model cards) explaining that complaints are triaged by an automated system, and how to escalate to a human if the routing seems wrong.
- Periodically sample and review complaints written in Pidgin or colloquial English to catch systematic misclassification patterns before they erode trust.
- Treat complaint feedback and correction requests as a lightweight, ongoing signal into model evaluation — not just a support ticket to close.

Business / Support Leadership — Manage Closely

This group holds budget and deployment authority, so engagement needs to be frequent, detailed, and framed around the metrics they are accountable for. Concrete actions:

- Deliver regular reporting on precision, recall, and F1-scores per complaint category, tied explicitly to triage-time reduction.
- Hold structured go/no-go review checkpoints before each deployment stage, rather than a single end-of-project sign-off.

- Surface known model limitations (e.g. weaker performance on underrepresented dialects or rare complaint classes) proactively, rather than only in a post-incident report.
- Involve leadership in setting the acceptable error-rate threshold up front, so evaluation criteria reflect what they actually consider deployment-ready.

References

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- [2] Birhane, A. et al. (2022). Power to the People? Opportunities and Challenges for Participatory AI. In *Proceedings of the 2nd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO '22)* (pp. 1–8). Association for Computing Machinery. <https://doi.org/10.1145/3551624.3555290>
- [3] Wakunuma, K., & Jiya, T. (2019). Stakeholder Engagement and Responsible Research & Innovation in Promoting Sustainable Development and Empowerment through ICT. *European Journal of Sustainable Development*, 8(3), 275–275. <https://doi.org/10.14207/ejsd.2019.v8n3p275>